

Principles of Programming Languages

Lecture 1: Introduction

Andrei Arusoaie¹

¹Department of Computer Science

October 3, 2019

Outline

General course information

Preliminary discussion

History

Main Paradigms

Principles of Programming Languages - Motivation

- ▶ Learning a new PL is nowadays a requirement
- ▶ “Programmers need to move from one PL to another with naturalness and speed” [Gabbrielli2010]
- ▶ PLs have their similarities, analogies, inherited characteristics

GOAL:

understand the basic mechanisms behind the creation of a PL

Principles of Programming Languages - Motivation

- ▶ Learning a new PL is nowadays a requirement
- ▶ “Programmers need to move from one PL to another with naturalness and speed” [Gabbrielli2010]
- ▶ PLs have their similarities, analogies, inherited characteristics

GOAL:

understand the basic mechanisms behind the creation of a PL

Principles of Programming Languages - Motivation

- ▶ Learning a new PL is nowadays a requirement
- ▶ “Programmers need to move from one PL to another with naturalness and speed” [Gabbrielli2010]
- ▶ PLs have their similarities, analogies, inherited characteristics

GOAL:

understand the basic mechanisms behind the creation of a PL

Principles of Programming Languages - Motivation

- ▶ Learning a new PL is nowadays a requirement
- ▶ “Programmers need to move from one PL to another with naturalness and speed” [Gabbrielli2010]
- ▶ PLs have their similarities, analogies, inherited characteristics

GOAL:

understand the basic mechanisms behind the creation of a PL

Skills to develop

- ▶ The definition of a programming language:

Syntax + Semantics

- ▶ Have a better understanding of the PLs you already know
- ▶ Ability to learn a new PL fast
- ▶ Get familiar with a framework for defining languages

Skills to develop

- ▶ The definition of a programming language:

Syntax + Semantics

- ▶ Have a better understanding of the PLs you already know
- ▶ Ability to learn a new PL fast
- ▶ Get familiar with a framework for defining languages

Skills to develop

- ▶ The definition of a programming language:

Syntax + Semantics

- ▶ Have a better understanding of the PLs you already know
- ▶ Ability to learn a new PL fast
- ▶ Get familiar with a framework for defining languages

Skills to develop

- ▶ The definition of a programming language:

Syntax + Semantics

- ▶ Have a better understanding of the PLs you already know
- ▶ Ability to learn a new PL fast
- ▶ Get familiar with a framework for defining languages

Principles of Programming Languages - Organisation

- ▶ People: Andrei Arusoaie and Andreea Buterchi
- ▶ Period: 1st semester (Fall 2019), 2nd year
- ▶ <http://www.uaic.ro/wp-content/uploads/2019/05/structura-an-universitar.pdf>
- ▶ 14 (full activity) weeks
- ▶ Final grade = 40 (midterm) + 40 (final exam) + 20 (lab)
- ▶ Midterm: 40 points, week 8 - written test
- ▶ Project: 40 points, week 14 - written test
- ▶ Lab: 10 points for attending the lab
- ▶ Lab: 10 points for lab activity + bonuses (!max 20 points)
- ▶ Bonuses: presentations/interesting projects/etc.
- ▶ Criteria: min 40 points exam + min 10 points lab

Principles of Programming Languages - Organisation

- ▶ People: Andrei Arusoaie and Andreea Buterchi
- ▶ Period: 1st semester (Fall 2019), 2nd year
- ▶ <http://www.uaic.ro/wp-content/uploads/2019/05/structura-an-universitar.pdf>
- ▶ 14 (full activity) weeks
- ▶ Final grade = 40 (midterm) + 40 (final exam) + 20 (lab)
- ▶ Midterm: 40 points, week 8 - written test
- ▶ Project: 40 points, week 14 - written test
- ▶ Lab: 10 points for attending the lab
- ▶ Lab: 10 points for lab activity + bonuses (!max 20 points)
- ▶ Bonuses: presentations/interesting projects/etc.
- ▶ Criteria: min 40 points exam + min 10 points lab

Principles of Programming Languages - Organisation

- ▶ People: Andrei Arusoaie and Andreea Buterchi
- ▶ Period: 1st semester (Fall 2019), 2nd year
- ▶ <http://www.uaic.ro/wp-content/uploads/2019/05/structura-an-universitar.pdf>
- ▶ 14 (full activity) weeks
- ▶ Final grade = 40 (midterm) + 40 (final exam) + 20 (lab)
- ▶ Midterm: 40 points, week 8 - written test
- ▶ Project: 40 points, week 14 - written test
- ▶ Lab: 10 points for attending the lab
- ▶ Lab: 10 points for lab activity + bonuses (!max 20 points)
- ▶ Bonuses: presentations/interesting projects/etc.
- ▶ Criteria: min 40 points exam + min 10 points lab

Principles of Programming Languages

Course description:

- ▶ RO:

`https://profs.info.uaic.ro/~arusoae.andrei/lectures/PLP/2019/fisa\_disciplinei\_plp.pdf`

- ▶ EN: `https://profs.info.uaic.ro/~arusoae.andrei/lectures/PLP/2019/course\_description\_plp\_en.pdf`

Website:

- ▶ `https://profs.info.uaic.ro/~arusoae.andrei/lectures/PLP/2019/plp.html`

Bibliography

► Bibliography:

1. **Practical Foundations of Programming Languages**, Robert Harper

<https://www.cs.cmu.edu/~rwh/pfpl/2nded.pdf>

2. **Software Foundations** - Volume 2, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey

<https://softwarefoundations.cis.upenn.edu/plf-current/index.html>

3. **Programming Languages: Principles and Paradigms**, Maurizio Gabbrielli, Simone Martini

http://websrv.dthu.edu.vn/attachments/newsevents/content2415/Programming_Languages_-_Principles_and_Paradigms_thereds1106.pdf

Labs

- ▶ Lab: Coq
- ▶ Web: <https://coq.inria.fr/>
- ▶ Software Foundations - Volume 2, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey
<https://softwarefoundations.cis.upenn.edu/plf-current/index.html>

Labs

- ▶ Lab: Coq
- ▶ Web: `https://coq.inria.fr/`
- ▶ **Software Foundations** - Volume 2, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hrițcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey

`https://softwarefoundations.cis.upenn.edu/
plf-current/index.html`

Principles of Programming Languages - Communication

- ▶ **Course page:**

`https:`

`//profs.info.uaic.ro/~arusoae.andrei/lectures/PLP/2019/plp.html`

- ▶ **Email:** `arusoae.andrei@info.uaic.ro`

Today's lecture

1. Preliminary discussion
2. History
3. Programming paradigms
4. Debate & preparing lab

Today's lecture

1. Preliminary discussion
2. History
3. Programming paradigms
4. Debate & preparing lab

Today's lecture

1. Preliminary discussion
2. History
3. Programming paradigms
4. Debate & preparing lab

Today's lecture

1. Preliminary discussion
2. History
3. Programming paradigms
4. Debate & preparing lab

Quick questions, debatable answers

- ▶ What is a programming language?
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

Quick questions, debatable answers

- ▶ **What is a programming language?**
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

Quick questions, debatable answers

- ▶ What is a programming language?
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

Quick questions, debatable answers

- ▶ What is a programming language?
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

Quick questions, debatable answers

- ▶ What is a programming language?
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

Quick questions, debatable answers

- ▶ What is a programming language?
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

Quick questions, debatable answers

- ▶ What is a programming language?
- ▶ Why do we need programming languages?
- ▶ Which languages have you heard about?
- ▶ Have you ever created your own programming language?
- ▶ How to design a programming language?
- ▶ What is the best programming language?

History - I

- ▶ Programming computers
- ▶ 1944: ASCC/MARK I
 - ▶ built by IBM and Harvard Univ., H. Aiken
 - ▶ programmed using punched tapes and external physical media (like switches)
 - ▶ 10 m length, 4 tones, very expensive
- ▶ 1946: ENIAC
 - ▶ J. Mauchly and J.P. Eckert, early (design) J. von Neumann
 - ▶ Orders of magnitude faster than humans at computing
 - ▶ no storage
 - ▶ programmed using external physical media (electrical cables)
 - ▶ used to calculate ballistic trajectories

History - I

- ▶ Programming computers
- ▶ 1944: ASCC/MARK I
 - ▶ built by IBM and Harvard Univ., H. Aiken
 - ▶ programmed using punched tapes and external physical media (like switches)
 - ▶ 10 m length, 4 tones, very expensive
- ▶ 1946: ENIAC
 - ▶ J. Mauchly and J.P. Eckert, early (design) J. von Neumann
 - ▶ Orders of magnitude faster than humans at computing
 - ▶ no storage
 - ▶ programmed using external physical media (electrical cables)
 - ▶ used to calculate ballistic trajectories

History - II

- ▶ 1949: EDSAC - the first “computer” in the modern sense
 - ▶ electronic and digital
 - ▶ able to perform four elementary arithmetic operations
 - ▶ programable
 - ▶ storage of programs and data

History - III

- ▶ 1GL (first generation languages): machine language
 - ▶ elementary instructions immediately executable by a processor
 - ▶ coding: completely manual
 - ▶ hard to use, too “machine connected”
 - ▶ people realized that they need something closer to the user’s natural language
- ▶ 2GL: (second generation languages): assembly language
 - ▶ introduced to ease the development
 - ▶ symbolic representation of the machine language
 - ▶ translated into machine code
 - ▶ downside: each machine has its own assembly language

History - III

- ▶ 1GL (first generation languages): machine language
 - ▶ elementary instructions immediately executable by a processor
 - ▶ coding: completely manual
 - ▶ hard to use, too “machine connected”
 - ▶ people realized that they need something closer to the user’s natural language
- ▶ 2GL: (second generation languages): assembly language
 - ▶ introduced to ease the development
 - ▶ symbolic representation of the machine language
 - ▶ translated into machine code
 - ▶ downside: each machine has its own assembly language

History - IV

- ▶ 1950 – 3GL: high-level languages
 - ▶ abstract languages
 - ▶ ignore the physical characteristics of the computer
 - ▶ well suited to express *algorithms*
- ▶ 1950: ALGOL = ALGOritmic Languages (family of languages)
- ▶ 1957: FORTRAN = FORmula TRANslation
- ▶ 1960: LISP = LISt Processor
- ▶ However, the development was still an issue
- ▶ Lots of human resources required, expensive hardware

History - IV

- ▶ 1950 – 3GL: high-level languages
 - ▶ abstract languages
 - ▶ ignore the physical characteristics of the computer
 - ▶ well suited to express *algorithms*
- ▶ 1950: ALGOL = ALGOritmic Languages (family of languages)
- ▶ 1957: FORTRAN = FORmula TRANslation
- ▶ 1960: LISP = LISt Processor
- ▶ However, the development was still an issue
- ▶ Lots of human resources required, expensive hardware

History - IV

- ▶ 1950 – 3GL: high-level languages
 - ▶ abstract languages
 - ▶ ignore the physical characteristics of the computer
 - ▶ well suited to express *algorithms*
- ▶ 1950: ALGOL = ALGOritmic Languages (family of languages)
- ▶ 1957: FORTRAN = FORmula TRANslation
- ▶ 1960: LISP = LISt Processor
- ▶ However, the development was still an issue
- ▶ Lots of human resources required, expensive hardware

History - V

- ▶ 1970's: microprocessor; batch processing
- ▶ Further abstractions pushed PL development to what PL are today
- ▶ New paradigms: object-oriented, declarative programming
- ▶ The C language: Dennis Ritchie and Ken Thompson
 - ▶ Initially designed for the UNIX operating system
 - ▶ Successor of B
 - ▶ Followed by: Pascal, SmallTalk
- ▶ Declarative languages: ML (Meta Language), PROLOG

History - V

- ▶ 1970's: microprocessor; batch processing
- ▶ Further abstractions pushed PL development to what PL are today
- ▶ New paradigms: object-oriented, declarative programming
- ▶ The C language: Dennis Ritchie and Ken Thompson
 - ▶ Initially designed for the UNIX operating system
 - ▶ Successor of B
 - ▶ Followed by: Pascal, SmallTalk
- ▶ Declarative languages: ML (Meta Language), PROLOG

History - V

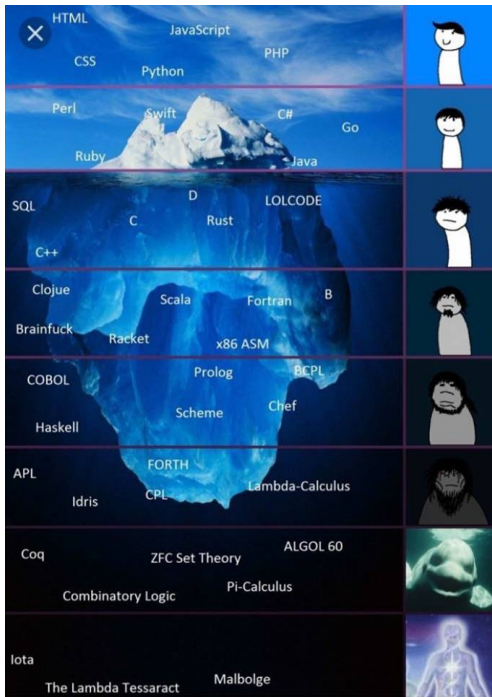
- ▶ 1970's: microprocessor; batch processing
- ▶ Further abstractions pushed PL development to what PL are today
- ▶ New paradigms: object-oriented, declarative programming
- ▶ The C language: Dennis Ritchie and Ken Thompson
 - ▶ Initially designed for the UNIX operating system
 - ▶ Successor of B
 - ▶ Followed by: Pascal, SmallTalk
- ▶ Declarative languages: ML (Meta Language), PROLOG

History - VI

- ▶ 1980's: the PC era
- ▶ 1986: **C++** - Bjarne Stroustrup
- ▶ 1989: **HTML** - T. Berners-Lee, **Python** - Guido van Rossum
- ▶ 1990's: internet, WWW
- ▶ 1990-1995: **Java** - Jim Gosling at SUN
- ▶ 1994 : **PHP** - Rasmus Lerdorf
- ▶ 2000 : (Microsoft) **C#** - Anders Hejlsberg
- ▶ 2004 : **Scala** - Martin Odersky
- ▶ 2012 : **Go** - at Google Inc. (started in 2007 by R. Griesemer, R. Pike, K. Thompson)
- ▶ 2014 : **Swift** - at Apple Inc. (started in 2010 by Chris Lattner)

History - VI

- ▶ 1980's: the PC era
- ▶ 1986: **C++** - Bjarne Stroustrup
- ▶ 1989: **HTML** - T. Berners-Lee, **Python** - Guido van Rossum
- ▶ 1990's: internet, WWW
- ▶ 1990-1995: **Java** - Jim Gosling at SUN
- ▶ 1994 : **PHP** - Rasmus Lerdorf
- ▶ 2000 : (Microsoft) **C#** - Anders Hejlsberg
- ▶ 2004 : **Scala** - Martin Odersky
- ▶ 2012 : **Go** - at Google Inc. (started in 2007 by R. Griesemer, R. Pike, K. Thompson)
- ▶ 2014 : **Swift** - at Apple Inc. (started in 2010 by Chris Lattner)



Main programming paradigms

1. Imperative programming
 - ▶ *First do this and then do that*
2. Object-oriented programming
 - ▶ *Model the world using objects that exchange messages*
3. Functional programming¹
 - ▶ *Evaluate an expression and pass the result*
4. Logic programming¹
 - ▶ *Answer questions by searching a solution*

¹Part of the **Declarative** programming paradigm

Main programming paradigms

1. Imperative programming
 - ▶ *First do this and then do that*
2. Object-oriented programming
 - ▶ *Model the world using objects that exchange messages*
3. Functional programming¹
 - ▶ *Evaluate an expression and pass the result*
4. Logic programming¹
 - ▶ *Answer questions by searching a solution*

¹Part of the **Declarative** programming paradigm

Main programming paradigms

1. Imperative programming
 - ▶ *First do this and then do that*
2. Object-oriented programming
 - ▶ *Model the world using objects that exchange messages*
3. Functional programming¹
 - ▶ *Evaluate an expression and pass the result*
4. Logic programming¹
 - ▶ *Answer questions by searching a solution*

¹Part of the **Declarative** programming paradigm

Main programming paradigms

1. Imperative programming
 - ▶ *First do this and then do that*
2. Object-oriented programming
 - ▶ *Model the world using objects that exchange messages*
3. Functional programming¹
 - ▶ *Evaluate an expression and pass the result*
4. Logic programming¹
 - ▶ *Answer questions by searching a solution*

¹Part of the **Declarative** programming paradigm

Imperative programming

Example program:

```
read(n);  
s = 0;  
if (n < 0) {  
    print "error";  
} else {  
    while (n > 0) do {  
        s = s + n;  
        n = n - 1;  
    }  
}
```

- ▶ Requires a *program state*
- ▶ Execution is (incrementally) changing the program state
- ▶ Instructions: assignment, decisional, loops
- ▶ *First class value*: variable

Imperative programming

Example program:

```
read(n);  
s = 0;  
if (n < 0) {  
    print "error";  
} else {  
    while (n > 0) do {  
        s = s + n;  
        n = n - 1;  
    }  
}
```

- ▶ Requires a *program state*
- ▶ Execution is (incrementally) changing the program state
- ▶ Instructions: assignment, decisional, loops
- ▶ *First class value*: variable

Imperative programming

Example program:

```
read(n);  
s = 0;  
if (n < 0) {  
    print "error";  
} else {  
    while (n > 0) do {  
        s = s + n;  
        n = n - 1;  
    }  
}
```

- ▶ Requires a *program state*
- ▶ Execution is (incrementally) changing the program state
- ▶ Instructions: assignment, decisional, loops
- ▶ *First class value*: variable

Imperative programming

Example program:

```
read(n);  
s = 0;  
if (n < 0) {  
    print "error";  
} else {  
    while (n > 0) do {  
        s = s + n;  
        n = n - 1;  
    }  
}
```

- ▶ Requires a *program state*
- ▶ Execution is (incrementally) changing the program state
- ▶ Instructions: assignment, decisional, loops
- ▶ *First class value*: variable

Imperative programming

Example program:

```
read(n);  
s = 0;  
if (n < 0) {  
    print "error";  
} else {  
    while (n > 0) do {  
        s = s + n;  
        n = n - 1;  
    }  
}
```

- ▶ Requires a *program state*
- ▶ Execution is (incrementally) changing the program state
- ▶ Instructions: assignment, decisional, loops
- ▶ *First class value*: variable

Object-Oriented Programming

Example (Java):

```
public class Person {  
    private String name;  
    Person(String name) {  
        this.name = name;  
    }  
    public String getName() {  
        return name;  
    }  
}  
  
// in the main function  
Driver john = new Driver("John");  
john.startDriving();  
boolean status = john.getStatus();
```

```
public class Driver extends Person  
{  
    private boolean isDriving;  
    Driver(String name) {  
        super(name);  
        isDriving = false;  
    }  
    public void startDriving() {  
        isDriving = true;  
    }  
    public boolean getStatus() {  
        return isDriving;  
    }  
}
```

Object-Oriented Programming

- ▶ Model using *objects* and *classes*
- ▶ Objects exchange messages (dispatch/message passing)
- ▶ Abstraction, Encapsulation
- ▶ Inheritance, Polymorphism, Overriding, Overloading

Functional programming

Example: (Haskell)

```
mysum :: Integer -> Integer
mysum n = if (n < 0)
           then 0
           else n + mysum (n - 1)
```

Run:

```
> mysum 10
55
> mysum 0
0
> mysum (-3)
0
```

- ▶ Computations = evaluation of mathematical functions
- ▶ No state!
- ▶ *Non-mutable* values
- ▶ First class value: function

Functional programming

Example: (Haskell)

```
mysum :: Integer -> Integer
mysum n = if (n < 0)
           then 0
           else n + mysum (n - 1)
```

Run:

```
> mysum 10
55
> mysum 0
0
> mysum (-3)
0
```

- ▶ Computations = evaluation of mathematical functions
- ▶ No state!
- ▶ *Non-mutable* values
- ▶ First class value: function

Functional programming

Example: (Haskell)

```
mysum :: Integer -> Integer
mysum n = if (n < 0)
           then 0
           else n + mysum (n - 1)
```

Run:

```
> mysum 10
55
> mysum 0
0
> mysum (-3)
0
```

- ▶ Computations = evaluation of mathematical functions
- ▶ No state!
- ▶ *Non-mutable* values
- ▶ First class value: function

Logic Programming

Example (Prolog):

```
man(john).  
man(dan).  
man(mark).  
father(john, mark).  
father(mark, dan).  
  
grandfather(X,Y) :- man(X),  
man(Y), father(X,Z), father(Z,Y).
```

Search solution:

```
1 ?- father(mark, dan).  
true.  
2 ?- grandfather(mark, dan).  
false.  
3 ?- grandfather(john, dan).  
true.  
4 ?- grandfather(X, dan).  
X = john .
```

- ▶ Algorithm = Logic + Control
- ▶ Programmer provides a logical specification
- ▶ An interpreter searches for the solution using **resolution**

Logic Programming

Example (Prolog):

```
man(john).  
man(dan).  
man(mark).  
father(john, mark).  
father(mark, dan).  
  
grandfather(X,Y) :- man(X),  
man(Y), father(X,Z), father(Z,Y).
```

Search solution:

```
1 ?- father(mark, dan).  
true.  
2 ?- grandfather(mark, dan).  
false.  
3 ?- grandfather(john, dan).  
true.  
4 ?- grandfather(X, dan).  
X = john .
```

- ▶ Algorithm = Logic + Control
- ▶ Programmer provides a logical specification
- ▶ An interpreter searches for the solution using **resolution**

Logic Programming

Example (Prolog):

```
man(john).  
man(dan).  
man(mark).  
father(john, mark).  
father(mark, dan).  
  
grandfather(X,Y) :- man(X),  
man(Y), father(X,Z), father(Z,Y).
```

Search solution:

```
1 ?- father(mark, dan).  
true.  
2 ?- grandfather(mark, dan).  
false.  
3 ?- grandfather(john, dan).  
true.  
4 ?- grandfather(X, dan).  
X = john .
```

- ▶ Algorithm = Logic + Control
- ▶ Programmer provides a logical specification
- ▶ An interpreter searches for the solution using **resolution**

Domain Specific Languages

- ▶ Web: HTML, CSS, scripting (e.g., Javascript, PHP)
- ▶ Databases: SQL
- ▶ Algebraic: Maude, CafeOBJ
- ▶ Modelling: UML
- ▶ WSC: BPEL
- ▶ Regex text processing: Perl
- ▶ Programming languages: Rascal, Spoofax, **Coq**, Racket, K, ...
- ▶ ...

Debate

Which paradigm is better?

Indeed, this question is stupid...

Debate

Which paradigm is better?

Indeed, this question is stupid...

What's next?

- ▶ **PL = Syntax + Semantics**
- ▶ Syntax:
 - ▶ how can we combine symbols to create correct programs?
 - ▶ Grammars (BNF = Backus-Naur Form)
- ▶ Semantics: what programs *mean*?
- ▶ Other things related to PLs: pragmatics, compilers, etc.

What's next?

- ▶ PL = Syntax + Semantics
- ▶ Syntax:
 - ▶ how can we combine symbols to create correct programs?
 - ▶ Grammars (BNF = Backus-Naur Form)
- ▶ Semantics: what programs *mean*?
- ▶ Other things related to PLs: pragmatics, compilers, etc.

What's next?

- ▶ PL = Syntax + Semantics
- ▶ Syntax:
 - ▶ how can we combine symbols to create correct programs?
 - ▶ Grammars (BNF = Backus-Naur Form)
- ▶ Semantics: what programs *mean*?
- ▶ Other things related to PLs: pragmatics, compilers, etc.

What's next?

- ▶ PL = Syntax + Semantics
- ▶ Syntax:
 - ▶ how can we combine symbols to create correct programs?
 - ▶ Grammars (BNF = Backus-Naur Form)
- ▶ Semantics: what programs *mean*?
- ▶ Other things related to PLs: pragmatics, compilers, etc.

Questions?

Bibliography:

- ▶ *Chapter 13 from Programming Languages: Principles and Paradigms*, Maurizio Gabbrielli, Simone Martini, Online:
http://websrv.dthu.edu.vn/attachments/newsevents/content2415/Programming_Languages_-_Principles_and_Paradigms_thereds1106.pdf